
Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Sleiman

1. Purpose & Objective

The purpose of this document is to establish a structured reporting standard for the duration of the internship. This framework is designed to track progress and bridge the gap between Eaton's advanced technologies and local market needs, specifically focusing on *Albonian Group*'s commercial distribution and *Technicat Group*'s panel manufacturing capabilities.

2. Daily Report Components

Every daily report submitted will follow a standardized four-part structure to ensure consistency and readability.

Part I: Executive Summary

A concise, general overview of the day's primary focus. This section serves as a quick status check, summarizing the key technologies engaged with and the operational context.

Part II: Technical Learnings & Activities

A detailed engineering log summarizing the specific automation and software skills acquired.

- **Scope:** PLC Programming (easySmart/CODESYS), Communication Protocols (Modbus/TCP), and Digital Twin Design and Simulation.

Part III: Training Evaluation & Delivery Feedback

An honest evaluation of the training session itself, as requested.

- **Delivery Method:** Assessment of how the concepts were conveyed (e.g., Self-Study, Engineer Shadowing, Software Simulation).
- **Feedback:** Constructive suggestions on how the software training modules could be improved or optimized for future interns.

Part IV: Strategic Observations & Recommendations

A dual-focus analysis connecting Eaton Dubai's operations to the specific needs of the Lebanese partners:

- **For Albonian Group (Commercial):** Identification of new Eaton products, software features, or unique selling points observed in Dubai that should be introduced to the Lebanese market.
- **For Technicat Group (Manufacturing):** Observations on panel assembly techniques, and smart devices that can be implemented to improve production quality and efficiency.

3. Submission

- **Frequency:** Daily (End of Business Day).
- **Format:** Professional LaTeX generated PDF.
- **Recipient:** Mr. Zokhrof Slieman (Albonian Group).

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 01

December 17, 2025

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Today was my first day at the Eaton office which started with a tour of the site. I was supposed to directly get handed a laptop with the required software to directly jump into the learning process of automation programming. Unfortunately, my account was not ready yet so I spent my day researching and reviewing literature about a few IEC standard documentations.

Part II: Technical Learnings & Activities

- **LV Systems & IEC 61439:** Studied the distinction between Switchboards (Pow-R-Line CS) and Switchgear (Pow-R-Line C).
- **Key Insight (Form 4b):** Unlike Form 2 (common in simpler builds), Form 4b physically compartmentalizes the Busbar, Functional Unit, and Terminals. This is critical for Data Centers as it allows maintenance on one feeder while the rest of the board remains live.
- **Smart Protection (DigiTrip):** Analyzed the shift from thermal-magnetic MCCBs to electronic solid-state units.
- **Key Insight:** DigiTrip units act as data servers. They provide "Cause of Trip" LEDs and metering data, essential for troubleshooting industrial faults.
- **Automation Protocols (Modbus):** Reviewed the data exchange model for integrating breakers with PLCs.
- **Key Insight:** Eaton devices use Modbus RTU (Serial) internally and Modbus TCP (Ethernet) for system integration.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method:** Ryan Tabbara, Application Engineer, did a great job at introducing briefly the different software used. Due to the lack of internet access today, training was limited to documentation review rather than software simulation.
- **Feedback/Improvement:** The theoretical literature provided was comprehensive. However, finalizing the my user account is critical to enable access to "EasySoft" and "eBID" tomorrow, as hands-on simulation is the most effective way to apply these logic concepts.

Part IV: Strategic Observations & Recommendations

- **For Albonian Group (Commercial):**
 - **Selling Safety:** When pitching to critical buildings such as hospitals, we should emphasize the **maintenance safety** of Form 4b (live work capability) rather than just price. This is a major differentiator against lower-tier competitors.
- **For Technicat Group (Manufacturing):**
 - **Error Prevention (eBID):** Observed the use of "Eaton Bid Manager." This tool automatically forces the inclusion of specific copper kits and barriers when "Form 4b" is selected. Adopting similar validation logic in our quoting would prevent BOM errors in our complex panel projects.
 - **IP Ratings:** Noticed strict adherence to IEC 60529 (IP54 min) for industrial panels here. We should prepare for higher IP requirements if we target Gulf-standard projects due to their nature of the environment.

Plan for Tomorrow

- Receive my Eaton account and devices.
- Visit with Mr. Martin to Eaton's Customer Experience Center.
- Introduction to EasySoft for basic logic programming.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Days 02 & 03

Dec 18-19, 2025

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Due to severe weather conditions and government safety mandates, operations shifted to remote work midway through Day 2 and continued through Day 3. I am still waiting from IT to solve the issue with my user (scheduled for Monday). I maximized productivity by visiting the Eaton Customer Experience Center (CEC) to study Eaton's various product line, learned about the "BreakerVisu" system and Ladder Logic fundamentals.

Part II: Technical Learnings & Activities

- **Customer Experience Center (CEC) Visit:** Toured the demo facility featuring live VFD, UPS, and ATS setups.
- **Key Insight (Oil & Gas):** Observed the drastic difference in build quality for O&G switchgear. The safety interlocking and explosion-proof requirements create a much higher complexity level compared to standard commercial panels. Seeing how cooling is done through intelligent spacing inside the enclosure rather than relying on conventional cooling techniques.
- **Hardware Study - BreakerVisu:** Analyzed the technical manual for the Eaton BreakerVisu.
- **Zero-Programming Config:** Unlike a standard HMI/PLC, BreakerVisu auto-detects connected breakers. It logs energy consumption and trip events automatically without writing code.
- **Automation Fundamentals (Remote Study):** Since I could not access EasySoft yet, I practiced designing Ladder Diagrams (LD) on paper.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method: Hybrid (On-Site Demo)** The visit to the CEC was extremely beneficial, especially with having Mr. Melbin, a senior engineer at Eaton, explaining thoroughly each product and answering questions with clarity.
- **Feedback/Improvement:** The weather disruption was unavoidable. However, the delay with IT has now consumed three days excluding the weekend.

Part IV: Strategic Observations & Recommendations

- **For Albonian Group (Commercial):**
 - **Strategic Pivot (Lebanon Offshore Gas):** With the increasing international activity regarding Lebanon's offshore gas exploration, there will be a surge in demand for O&G systems. Eaton is a market leader in this niche sector. Albonian have a serious opportunity at hand to capitalize on.
 - **BreakerVisu Retrofit:** Since BreakerVisu supports Modbus RTU, we can sell this HMI to clients who *already* have non-Eaton breakers but want to add "Smart Monitoring" without replacing their whole panel.
- **For Technicat Group (Manufacturing):**
 - **The "Smart Panel" Upsell:** The BreakerVisu system requires **zero programming**. Technicat could offer a "Digital Panel" option to customers by simply installing this unit and connecting via SmartWire-DT. This adds high value (energy logging/event history) with almost zero additional engineering hours compared to a full PLC setup.

Plan for Monday (Post-Weekend)

- Have everything setup, from hardware to software.
- Translate paper-based Ladder Logic exercises into actual simulation.
- Begin first automation project which will be assigned after finaling the setup.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 04

December 22, 2025

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

My day started with a training with Mr. Melbin about VFDs, their different applications & wiring. I finally got access to the demo version of Eaton's automation programming software "easySoft" on my own laptop. I will receive the full version once I get handed my laptop.

Part II: Technical Learnings & Activities

- **Variable Frequency Drive (Training with Mr. Mebin):**
 - **Control Schemes:** Learned the wiring distinction between **Local Control** (Potentiometer/Keypad on panel) and **Remote Control** (0-10V analog signals/Modbus from).
 - **Application:** Discussed the different parameters that need to be defined and fine tuned for optimal and desired performance.
- **PLC Programming (First Project - easySoft):**
 - **Logic Implementation:** Developed a Duty/Standby Pump Control System using Function Block Diagram (FBD). The logic ensures that if the active pump faults, the standby pump takes over immediately (Automatic Changeover).
 - **Hysteresis:** Programmed an analog comparator with a time delay to create a "Deadzone." This prevents the pumps from chattering (turning on/off rapidly) when the water level ripples near the sensor setpoint.
 - **Fault Management:** Integrated a specific alarm block over the previously built system that triggers if the active pump fails, signalling the changeover.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method:** Mr. Mebin's explanation of VFD setup and wiring was extremely clear and relevant to what we saw earlier in our visit to the CEC. The transition to coding on my own is an essential step to understand the basis of automation which will be further complemented by training from different professionals.
- **Feedback:** Mr. Melbin's explanation of VFDs showed his clear expertise about the topic.

Part IV: Strategic Observations & Recommendations

- **For Albonian Group (Commercial):**
 - **VFD Upsell:** Mr. Mebin highlighted that VFDs have their own electrical safety built within them and that they are not just for speed control but for mechanical protection (anti-jamming, pipe-fill mode). These are few of the selling points that make Eaton one of the largest shareholders in the VFD sector.
- **For Technicat Group (Manufacturing):**
 - **Standardized Logic Libraries:** The "Duty/Standby" logic I wrote today is a universal requirement for almost all buildings (Water/Diesel pumps).
 - **Action:** Technicat could create a "Standard Logic Library" for Eaton easyE4 controllers. Instead of rewriting code for every project, we load a pre-tested "Pump Script." This reduces engineering time on the shop floor and guarantees zero bugs during implementation.

Plan for Tomorrow

- In-person training session on advanced features of easySoft.
- Assignment of the Main Automation Project (Scheduled duration: Dec 23 - Jan 4).

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 05

December 23, 2025

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Focus: Triplex Booster Logic Design

Part I: Executive Summary

Today I was assigned the primary automation capstone project: designing the control logic for a Triplex Pressure Booster System (Duty, Standby, Assist). The day was dedicated to architecting the Function Block Diagram (FBD) to handle analog pressure thresholds, automatic alternation (wear leveling), and failover redundancy.

Part II: Technical Learnings & Activities

- **Project Scope (Triplex Booster System):**
 - **Configuration:** 3 Pumps (Duty, Standby, Assist) controlled via 0-16 bar analog pressure sensor.
 - **Parameters:** Cut-in pressure set at 3.5 bar; Cut-off set at 6 bar.
- **Logic Implementation (easySoft FBD):**
 - **Cascading Control:** Programmed the logic to trigger the *Duty* pump immediately when pressure drops below 3.5 bar. If demand remains high, the *Assist* pump engages after a timer delay to boost pressure.
 - **Wear Leveling (Cyclic Alternation):** Implemented a "Cycle Changeover" logic. Every time the system reaches the cut-off pressure (6 bar), the Duty and Standby pumps swap roles. This ensures both pumps age equally, preventing one from seizing due to inactivity.
 - **Forced Changeover:** Added a runtime timer to force a swap between Duty and Standby if the pumps run continuously for too long without reaching the cut-off pressure.
 - **Fault Redundancy:** Programmed "Instant Failover." If the Duty pump trips (Overload/Error), the Standby pump bypasses the timer and starts immediately to maintain line pressure.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method: Project-Based Learning.** Mr. Ryan Tabbara provided a detailed technical specification sheet rather than step-by-step instructions.
- **Feedback:** This workflow is excellent as it challenged me to interpret the client requirements and translate them into logic and block diagrams.

Plan for Tomorrow

- Develop "User Defined Function Blocks" (UDF) to make the code cleaner.
- Integrate the Analog Signal Simulation (Slider) to test the cut-in/cut-off points visually.
- Finalize the interlocks for the "Assist" pump fault handling.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Days 06 - 09

Dec 24,26,28,29, 2025

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

For the past 4 days I spent my time entirely working on the 3 Pump configuration system (DSA). Focusing on implementing the logic into FBD form, designing the PLC's built in HMI, and running simulations to test edge cases to ensure the robustness of the code.

Part II: Technical Learnings & Activities

• Logic Validation (Simulation Mode):

- **Edge Case Testing:** Simulated extreme scenarios to ensure system stability. Made sure the code/logic is neat to eliminate any discrepancies that might occur in some extreme scenarios and the system would continue normal operation.
- **Diagram:** Getting comfortable with the SLD shared by the client to understand the environment of the PLC with all the inputs/sensors and the expected outputs are very essential to ensure the logic is physically feasible.

• HMI & User Inputs (easyE4 Screen Configuration):

- **Linear Scaling:** Implemented the LS function block to map the raw 0-10V analog sensor signal to a human-readable 0-16 Bar value on the display.
- **User-Defined Inputs:** Created a menu system allowing the client/ facility manager to adjust the "Cut-In" and "Cut-Off" pressures as well as the Short and Long Delay Timers' duration directly from the PLC screen keypads, eliminating the need for a professional's intervention to change set points.
- **Text Displays:** Configured screen alerts to show real-time status of the system.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method:** Mr. Ryan Tabbara & I sat down and compared his code (actual code sent to the client) to mine and saw how each system reacted to different scenarios. By my final iteration our codes were acting the same but done through the use of different functional blocks.
- **Feedback:** Learning easySoft is tricky as there is no official manual/tutorial to guide a new student through the learning process. When stuck I had to rely on asking a professional as I could not find any detailed documentation online that could assist me.

Plan for Tomorrow

- Get assigned and start with Project #2

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Days 10, 11, 12, & 13

Dec 30, 2025 - Jan 05, 2026

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Over the past three working days, I executed and finalized "Project 2": a Chilled Water Pump Control System. Unlike the previous booster project, this required the logic to manage the simultaneous operation of two pumps with a load balancing feature. The project is fully simulated, validated, and the HMI interface is configured.

Part II: Technical Learnings & Activities

- **Project Scope (HVAC Chilled Water):**
 - **Requirement:** Control 3 Pumps where 2 must always run simultaneously (2 Duty / 1 Standby).
 - **Cyclic Rotation:** Implemented logic to rotate the active pair ($P1+P2 \rightarrow P2+P3 \rightarrow P3+P1$) to ensure even mechanical wear across all units.
- **Advanced Logic Implementation:**
 - **State Management:** Unlike simple on/off logic, I implemented a counter mechanism to define specific system states.
 - **Fault Redundancy:** If one of the two active pumps faults, the logic immediately triggers the third (Standby) pump to maintain the flow rate, ensuring no loss of cooling capacity.
- **HMI Configuration (PLC Display):**
 - **Visual Status:** Configured the Text Display block to show the current pump status and cycle running time.
 - **VFD Integration:** The usage of VFD does not affect the logic in anyway. VFD parameters are control are configured within the VFD itself. All the PLC does is give the enable signal to the VFD which then runs the pumps.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method: Independent Problem Solving.** I was given the prompt and left to design the architecture from scratch.
- **Feedback:** This project highlighted the difference between Booster logic (Pressure Thresholds) and HVAC logic (Continuous Sequence). Using the easySoft simulator was critical to verify that the transition between pump pairs (states) occurred smoothly without stopping the flow.

Plan for Next Day

- Meeting with Mr. Fadel Essaily (Supervisor) upon his return to the office.
- Training on Eaton's Life Safety products and services.
- Assignment of new tasks or commencement of the next training module.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 14

January 06, 2026

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Today, my project was reviewed and validated by Mr. Fadel. I have now advanced to the final project of my internship: a complex Generator Power Management System (PMS) utilizing Eaton's industrial-grade XC-200 PLC. This project shifts focus from simple logic to advanced System Integration, featuring RS-485 Modbus communication with power meters and Embedded Web Services for remote visualization.

Part II: Technical Learnings & Activities

- **Project Scope (Power Management System):**
 - **Logic:** Automatic Transfer Switch (ATS) logic prioritizing Utility power, with intelligent Load Shedding/Adding logic to sequence generators based on real-time demand.
 - **Hardware Upgrade:** Transitioned from the compact easyE4 to the **Eaton XC-200 Modular PLC**. This requires more advanced IP addressing.
- **Industrial Communication Protocols:**
 - **Modbus RTU (RS-485):** The project requires the PLC to communicate with an external Power Meter to read Voltage, Current, and KW data.
 - **Web Visualization:** I am tasked with enabling the PLC's **Embedded Web Server**. This allows the control interface to be accessed via a web browser (IP Communication), effectively turning the PLC into an IoT (Internet of Things) device.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method: Incentivized Challenge.** The project is due next Wednesday, but a target has been set to finish early to permit deployment on physical hardware.
- **Feedback:** This project is the perfect final challenge as it combines Logic (ATS), Communication (Modbus), and UI (Web). Moving to the XC-200 platform exposes me to the highest level of automation environment used in large infrastructure projects.

Part IV: Strategic Observations & Recommendations

- **For Technicat Group (Manufacturing):**
 - **The Remote Access of Data:** The Web Server feature on the XC-200 allows a Facility Manager to check the generator status from their smartphone (via VPN). Technicat should offer this as a "Premium Tier" panel option. It differentiates us from competitors who still force clients to walk down to the basement to check the generator status.

Plan for Tomorrow

- Visit the Customer Experience Center (CEC) for specialized training with Mr. Tony on the specific Power Meter model.
- Begin developing the Communication Driver logic to establish the link between the XC-200 PLC and the RS-485 Meter.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 15

January 07, 2026

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Today marked a major transition from the easyE4 smart relays to the industrial-grade Eaton XC-204 Modular PLC platform. I successfully architected the full software foundation for the "Peak Shaving" Power Management System using CODESYS. Key achievements include establishing the Data Unit Types (DUTs), configuring the Modbus RTU communication driver for energy metering, and mapping the XN300 Slice I/O hardware.

Part II: Technical Learnings & Activities

- **System Architecture (CODESYS Environment):**
 - **Structured Data:** Instead of using unstructured variables, I defined a **DUT (Data Unit Type)** named **ST_PowerMeter** to group Voltage, Current, and Power data. This ensures the code is scalable and professional.
 - **Global Variable Lists (GVL):** Segregated data into **GVL_Hardware** (Physical I/O) and **GVL_Web** (HMI/SCADA tags). This separation is critical for enabling the Web Visualizer features later.
- **Communication Hardware Integration:**
 - **Modbus RTU (RS-485):** Successfully configured the Modbus Master driver (19200 Baud, Even Parity).
 - **Object-Oriented Programming (OOP):** Created a **Function Block (FB)** named **FB_generator**. By "instantiating" this block twice (**fbGen1**, **fbGen2**), I can control multiple generators using a single code template, reducing programming errors.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method: Self-Guided Architectural Design.** The shift to CODESYS required a move from "Linear Programming" (Ladder) to "Continuous Function Chart" (CFC). I am constantly referring to different CoDeSys manuals shared online by companies like ABB & Topcon.
- **Feedback:** The capabilities of the XC-204 are exponentially higher than the easyE4. The ability to create "Function Blocks" (OOP) mirrors professional software development and is essential for managing the complexity of a Peak Shaving system involving multiple generators.

Plan for Tomorrow

- Open the **CFC Editor** to link the Data Structures to the Logic.
- Implement the **Hysteresis** blocks to prevent generator chattering during load fluctuations.
- Connect the Modbus data inputs to the **FB_generator** logic to finalize the automated start/stop sequence.

Daily Internship Report

Eaton Dubai — Winter 2025-26



Powering Business Worldwide

Day 16, 17, 18, & 19

January 8 - 13, 2026

Intern: Joe Sfeir

Supervisor: Mr. Zokhrof Slieman

Part I: Executive Summary

Today, I formally submitted the "Automated Power Management System" project. The system is fully validated, documented, and ready for deployment. The remainder of the day was dedicated to analyzing "Future Improvements," specifically how to extend this architecture to support Hybrid Solar PV integration and AI-driven load prediction.

Part II: Technical Learnings & Activities

- **Project Finalization:**
 - **Validation:** Confirmed that the "Peak Shaving" logic correctly identifies high-load scenarios and triggers the appropriate generator sequence.
 - **Documentation:** Compiled a comprehensive engineering report detailing the Hardware Configuration, Software Architecture, and Modbus Register Maps.
- **Future Technologies Research (Hybrid Systems):**
 - **Solar Co-Sharing:** Conceptualized logic to allow Solar PV Inverters to run in parallel with the Utility. This requires precise synchronization logic (Phase/Frequency matching) which the XC-204 can handle.
 - **AI Predictive Control:** Explored how historical data could be analyzed to *predict* power outages based on voltage fluctuations, allowing the generator to "Pre-Start" before the blackout hits.

Part III: Training Evaluation & Delivery Feedback

- **Delivery Method:** This project successfully bridged the gap between theoretical knowledge and industrial application.
- **Feedback:** The transition from EasySoft (Relay Logic) to CODESYS (Structured Text) has been the most valuable technical skill gained. It allows for handling complex data structures that simple relays cannot manage.

Part IV: Strategic Observations & Recommendations

- **For Albonian Group (Commercial):**
 - **The "Energy Crisis" Solution:** Given Lebanon's energy situation, pitching "Hybrid Solar Integration" is crucial. We should market the XC-204 not just as a Generator Controller, but as a "Microgrid Controller" capable of managing Utility, Diesel, and Solar simultaneously.
- **For Technicat Group (Manufacturing):**
 - **Data-Driven Maintenance:** I proposed an "Event Logging" feature that records every fault to a CSV file. Technicat could offer a service contract to clients where we analyze this data monthly to predict maintenance needs (e.g., "Generator 2 is taking 5 seconds longer to start – Battery check needed").

Plan for Tomorrow

- Final Internship Review with Management.
- Receivable of certificate from Eaton.
- Official conclusion of the Winter 2025-2026 Internship.